LESSON 5.1

Determine the Area of Parallelograms and Rhombuses

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a parallelogram using base × height.

WHY THIS MATTERS Your architecture firm is designing a parallelogram-shaped patio for a client's backyard. The client needs to know how many square feet of pavers to order. The patio has a base of 14 feet and a height of 9 feet.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Slanted side *Spanish: Lado inclinado*	A side that leans instead of going straight up and down. It is not the perpendicular height.	In a leaning parallelogram, use the dashed 90° height—not the slanted side—in A = b × h.
Parallelogram *Spanish: Paralelogramo*	A four-sided shape with two pairs of parallel sides.	opposite sides never meet
Parallel *Spanish: Paralelas*	Two lines or sides that stay the same distance apart and never meet.	The top and bottom of a parallelogram never get closer together — matching arrowheads (›) mark one parallel pair, double arrowheads (››) mark the other.
Base *Spanish: Base*	Any side of a parallelogram can be the base. Once you pick a base, the height must be measured perpendicular (at a 90° angle) to it.	b in A = b × h
Height *Spanish: Altura*	The straight-up distance from the base to the top.	h in A = b × h
Perpendicular *Spanish: Perpendicular*	Two lines that meet to make a square corner (90 degrees).	height ⊥ base

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Base 9 in., height 4 in., slanted side 5 in. Find the area.

1. Base = 9 in., height = 4 in.
2. Skip the 5 in. slanted side.
3. A = 9 × 4 = 36

Answer: 36 in²

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller parallelogram: base = 5 cm, height = 4 cm.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The base is _____ and the perpendicular height is _____ .
- I know that measure is the height because _____ .
- Area = _____ × _____ = _____ square units.
- The slanted side is not used because _____ .

Word bank parallelogram · base · height · perpendicular · square corner · area · formula · multiply · square units · slanted side · rectangle · rearrange

WATCH OUT!

A common mistake

The most common mistake in Area of Parallelograms is multiplying the base by the slanted side instead of the perpendicular height. For a parallelogram with a base of 16 ft, a perpendicular height of 9 ft, and a slanted side of 11 ft, students compute 16 × 11 = 176 square feet instead of the correct 16 × 9 = 144 square feet. The height must be the straight-up (perpendicular) distance between the base and its opposite side, not the length of the slanted side, even though the slanted side is often the number that 'looks' like it belongs in the formula.

LESSON 5.1

Determine the Area of Parallelograms and Rhombuses

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a parallelogram using base × height.

WHY THIS MATTERS Your architecture firm is designing a parallelogram-shaped patio for a client's backyard. The client needs to know how many square feet of pavers to order. The patio has a base of 14 feet and a height of 9 feet.

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Use the formula A = b × h to find the area of each parallelogram.

Parallelogram	Base	Height	Area
Patio X	6 ft	4 ft	
Patio Y	11 ft	3 ft	

HINT Each row gives a base and a height in feet — the empty Area column is what you fill in.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

What is the area of a parallelogram with base 10 cm and height 7 cm?

(A) 17 sq cm
(B) 34 sq cm
(C) 70 sq cm
(D) 70 cm

HINT You know both the base (10 cm) and the height (7 cm) — this is a direct use of the parallelogram area formula.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

A parallelogram patio on the firm's blueprint has an area of 54 sq ft and a base of 9 ft. What is the height?

(A) 6 ft
(B) 27 ft
(C) 45 ft
(D) 63 ft

HINT This time you're given the area (54 sq ft) and the base (9 ft) — the height is the missing piece.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

What is the area of a parallelogram with base 9 cm and height 5 cm?

(A) 14 sq cm

(B) 22.5 sq cm

(C) 45 sq cm

(D) 90 sq cm

HINT Spot the base (9 cm) and the height (5 cm). Which operation does the area formula use on them?

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

Which measurement is the height of a parallelogram?

(A) The length of the slanted side

(B) The perimeter divided by 4

(C) The perpendicular distance between the base and the opposite side

(D) The longest side

HINT Picture a leaning parallelogram. The height is not a side you can trace — it's a distance measured straight across.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Read the parallelogram** base = 7 ft, height = 4 ft
2. **Write the formula** A = base × height
3. **Substitute the values** A = 7 + 4
4. **Calculate the area** A = 11 sq ft

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Look at the substitution step. Did the student add the base and height, or multiply them?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

If you cut a triangle from one side of the parallelogram and move it to the other side, what shape do you get? How does this help you find the area?

> **SENTENCE STARTER** The area of the parallelogram equals base x height = ___ x ___ = ___ square feet, which is the same as a ___ with the same base and height.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.1

Determine the Area of Parallelograms and Rhombuses

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a parallelogram using base × height.

WHY THIS MATTERS Your architecture firm is designing a parallelogram-shaped patio for a client's backyard. The client needs to know how many square feet of pavers to order. The patio has a base of 14 feet and a height of 9 feet.

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table.

Parallelogram	Base	Height	Area
Patio A	14 ft	9 ft	
Patio B	8 ft		56 sq ft
Patio C		12 ft	96 sq ft
Patio D	15 ft	6 ft	

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each shape to its area.

_____ Triangle 8×6	(A) 24 sq units
_____ Parallelogram 5×4	(B) 20 sq units
_____ Rectangle 7×3	(C) 21 sq units
_____ Triangle 10×8	(D) 40 sq units
_____ Square side 6	(E) 36 sq units
_____ Parallelogram 9×2	(F) 18 sq units

3 **Write the name of the correct group on each line.**

Sort each parallelogram by whether its area is greater than 50 sq units or 50 or less.

Groups: **Area > 50 sq units** **Area ≤ 50 sq units**

_____ b = 12, h = 5

_____ b = 7, h = 6

_____ b = 10, h = 8

_____ b = 9, h = 4

_____ b = 11, h = 7

_____ b = 5, h = 10

4 **Mark and label each value on the number line.**

Find the missing height: b = 8, A = 48



SHOW YOUR THINKING

PART 2 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Read the parallelogram** base = 9 cm, slanted side = 10 cm, height = 6 cm
2. **Write the formula** A = base × height
3. **Substitute the values** A = 9 × 10
4. **Calculate the area** A = 90 sq cm

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

If you cut a triangle from one side of the parallelogram and move it to the other side, what shape do you get? How does this help you find the area?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.1

Determine the Area of Parallelograms and Rhombuses

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a parallelogram using base × height.

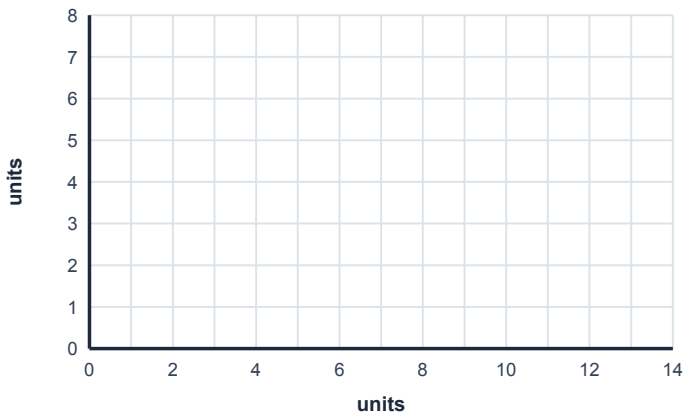
WHY THIS MATTERS Your architecture firm is designing a parallelogram-shaped patio for a client's backyard. The client needs to know how many square feet of pavers to order. The patio has a base of 14 feet and a height of 9 feet.

PART 1 Understand the idea

models and meaning

1 **Plot and label each point on the grid.**

Plot the parallelogram and find its area.



SHOW YOUR THINKING

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Area Error

1. **Identify base and height** b = 11 m, slant side = 8 m, h = 6 m
2. **Write the formula** A = b × h
3. **Substitute values** A = 11 × 8
4. **Calculate** A = 88 sq m

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A parallelogram has a base of 12 cm. If you double the height, the area doubles too. Explain why this happens using the formula, and give a specific example with numbers.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

If you cut a triangle from one side of the parallelogram and move it to the other side, what shape do you get? How does this help you find the area?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help